

SAT UZBEKISTAN

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1.

The manager of a gym selected a sample of 142 members at random to estimate the percentage of the gym's members that would continue to pay for a membership if the price increased. From the survey, the manager estimates that 89% of the gym's members would continue to pay for a membership if the price increased, with an associated margin of error of 5.15%. If the survey is repeated with a random sample of 284 members and the results are calculated in the same way, which of the following will be the most likely effect of using the larger random sample compared to the smaller random sample?

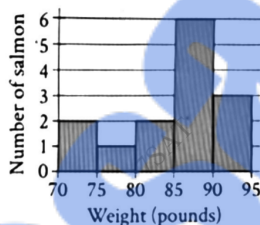
- ☐ (A) The margin of error will be lower.
- ☐ (B) The margin of error will be higher.
- ☐ (C) The estimate of the percentage of the gym's members that would continue to pay for a membership if the price increased will be lower.
- ☐ (D) The estimate of the percentage of the gym's members that would continue to pay for a membership if the price increased will be higher.

2.

A town manager selected 2 random samples of residents in the town to estimate the percentage of all residents that planned to vote in favor of a new business in the downtown area. Based on the first sample, he estimated that 75% of the residents in the town planned to vote in favor, with an associated margin of error of 8.5%. Based on the second sample, he estimated that 83% of the residents in the town planned to vote in favor, with an associated margin of error of 6.9%. Assuming the margins of error were calculated the same way, which of the following best explains why the first sample obtained a larger margin of error than the second sample?

- ☐ (A) The first sample contained more residents than the second sample.
- ☐ (B) The first sample contained fewer residents than the second sample.
- ☐ (C) The first sample contained a higher percentage of residents that planned to vote in favor of a new business in the downtown area.
- ☐ (D) The first sample contained a lower percentage of residents that planned to vote in favor of a new business in the downtown area.

3.



The histogram summarizes a data set of the weights, in pounds, of 14 salmon. If an additional weight of 16 pounds is added to the original data set to create a new data set of 15 weights, which of the following measures must be less for the new data set than for the original data set?

- I. The median
- II. The mean

- ☐ (A) I only
- ☐ (B) II only
- ☐ (C) I and II
- ☐ (D) Neither I nor II

4.

Each of the following frequency tables represents a data set. Which of these frequency tables represents the data set with the smallest standard deviation?

(A)

Value	Frequency
4	0
5	10
6	40
7	10
8	0

(B)

Value	Frequency
4	8
5	14
6	16
7	14
8	8

(C)

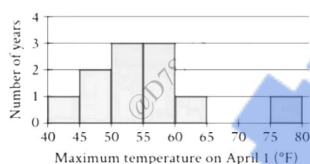
Value	Frequency
4	12
5	12
6	12
7	12
8	12

(D)

Value	Frequency
4	15
5	11
6	8
7	11
8	15

5.

The maximum temperature on April 1, in degrees Fahrenheit ($^{\circ}\text{F}$), was recorded each year at a certain weather station for 11 years. The histogram summarizes the recorded data set.



The temperature of 78.2°F is removed from this data set to create a new data set of 10 temperatures. Which of the following statements must be true?

- I. The mean of the new data set is less than the mean of the original data set.
- II. The median of the new data set is less than the median of the original data set.

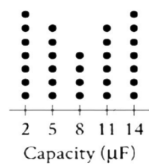
(A) I only

(B) II only

(C) I and II

(D) None

6.



The dot plot shows the distribution of capacity for a set of capacitors, set A , which a researcher used for a certain experiment. For another experiment, the researcher used a different set of capacitors, set B . Set B has the same number of capacitors as set A but the capacity of each capacitor of set B is 17 microfarads (μF) greater than the capacity of each respective capacitor in set A . Which of the following is true about the capacities of the capacitors in set B ?

(A) The mean capacity is $8\mu\text{F}$, and the range of capacities is $12\mu\text{F}$.

(B) The mean capacity is $8\mu\text{F}$, and the range of capacities is $29\mu\text{F}$.

(C) The mean capacity is $25\mu\text{F}$, and the range of capacities is $12\mu\text{F}$.

(D) The mean capacity is $25\mu\text{F}$, and the range of capacities is $29\mu\text{F}$.

7.

Each of the following frequency tables represents a data set. Which data set has the greatest mean?

(A)

Value	Frequency
60	2
70	3
80	4
90	5

(B)

Value	Frequency
60	6
70	3
80	3
90	6

(C)

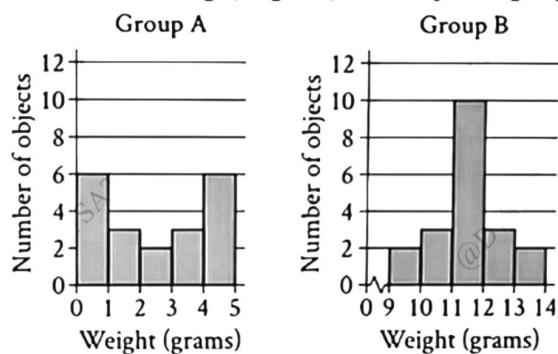
Value	Frequency
60	4
70	4
80	4
90	4

(D)

Value	Frequency
60	5
70	4
80	4
90	5

8.

The weight of each object in two groups, A and B, was recorded. The histograms represent the distributions of weight, in grams, for the objects in groups A and B.



The mean weight of the objects in group A is equal to 2.1, and the mean weight of the objects in group B is equal to 11.1. Which of the following statements about the standard deviations of weight for the objects in these two groups is true?

- (A) The standard deviation of weight for the objects in group A is less than the standard deviation of weight for the objects in group B.
- (B) The standard deviation of weight for the objects in group A is equal to the standard deviation of weight for the objects in group B.
- (C) The standard deviation of weight for the objects in group A is greater than the standard deviation of weight for the objects in group B.
- (D) There is not enough information to compare the standard deviations of weight for the objects in these two groups.

9.

Each of the following frequency tables represents a data set. Which of these frequency tables represents the data set with the smallest standard deviation?

(A)

Value	Frequency
3	0
4	10
5	55
6	10
7	0

(B)

Value	Frequency
3	11
4	17
5	19
6	17
7	11

(C)

Value	Frequency
3	15
4	15
5	15
6	15
7	15

(C)

Value	Frequency
3	15
4	15
5	15
6	15
7	15

(D)

Value	Frequency
3	17
4	15
5	11
6	15
7	17

10.

Each of the following frequency tables represents a data set. Which of these frequency tables represents the data set with the smallest standard deviation?

(A)

Value	Frequency
5	0
6	10
7	55
8	10
9	0

(B)

Value	Frequency
5	11
6	17
7	19
8	17
9	11

(C)

Value	Frequency
5	15
6	15
7	15
8	18
9	15

(D)

Value	Frequency
5	18
6	14
7	11
8	14
9	18

11.

A town manager selected 2 random samples of residents in the town to estimate the percentage of all residents that planned to vote in favor of a new business in the downtown area. Based on the first sample, he estimated that 75% of the residents in the town planned to vote in favor, with an associated margin of error of 8.5%. Based on the second sample, he estimated that 83% of the residents in the town planned to vote in favor, with an associated margin of error of 6.9%. Assuming the margins of error were calculated the same way, which of the following best explains why the first sample obtained a larger margin of error than the second sample?

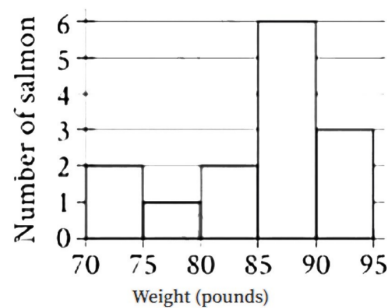
(A) The first sample contained more residents than the second sample.

(B) The first sample contained fewer residents than the second sample.

(C) The first sample contained a higher percentage of residents that planned to vote in favor of a new business in the downtown area.

(D) The first sample contained a lower percentage of residents that planned to vote in favor of a new business in the downtown area.

12.



The histogram summarizes a data set of the weights, in pounds, of 14 salmon. If an additional weight of 32 pounds is added to the original data set to create a new data set of 15 weights, which of the following measures must be less for the new data set than for the original data set?

- I. The median
- II. The mean

☐ A I only

☐ B II only

☐ C I and II

☐ D Neither I nor II

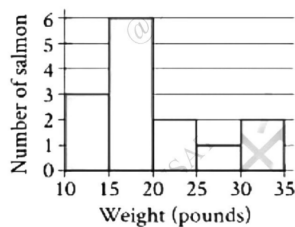
13.

Number of birds	Number of days
14	3
15	3
16	4
17	1
18	3
19	9

A science class studied the feeding behavior of the birds that visited a bird feeder each school day for a period of time. The table summarizes the number of birds the class observed visiting the bird feeder each school day during this period of time. What is the median number of birds the class observed visiting the bird feeder each school day during this period of time?

Answer Preview:

14.



The histogram summarizes a data set of the weights, in pounds, of 14 salmon. If an additional weight of 68 pounds is added to the original data set to create a new data set of 15 weights of salmon, which of the following measures must be greater for the new data set than for the original data set?

- I. The median
- II. The mean

- ☐ (A) I only
- ☐ (B) II only
- ☐ (C) I and II
- ☐ (D) Neither I nor II

15.

Data set A: 8, 11, 12, 15, 21, 27

Data set B: 9, 12, 13, 16, 22, h

Data sets A and B each consist of 6 values as shown, where h is a constant. If the standard deviation of data set A is greater than the standard deviation of data set B , which of the following could be the value of h ?

- I. 23
- II. 28
- III. 29

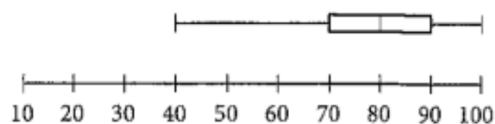
- ☐ (A) I only
- ☐ (B) II only
- ☐ (C) III only
- ☐ (D) II or III

16.

A research manager selected 2 random samples of ovens of a certain type to estimate the average amount of time this type of oven takes to preheat to 325 degrees Fahrenheit ($^{\circ}\text{F}$). The research manager recorded the amount of time, in minutes, each oven takes to preheat to 325 $^{\circ}\text{F}$. Based on the first sample, the research manager estimated that this type of oven takes an average of 15.2 minutes to preheat to 325 $^{\circ}\text{F}$, with an associated margin of error of 1 minute. Based on the second sample, the research manager estimated that this type of oven takes an average of 15.4 minutes to preheat to 325 $^{\circ}\text{F}$, with an associated margin of error of 2.2 minutes. Assuming the margins of error were calculated the same way, which of the following best explains why the first sample obtained a smaller margin of error than the second sample?

- ☐ (A) The first sample contained more ovens than the second sample.
- ☐ (B) The first sample contained fewer ovens than the second sample.
- ☐ (C) The first sample took more time on average to preheat to 325 $^{\circ}\text{F}$ than the second sample.
- ☐ (D) The first sample took less time on average to preheat to 325 $^{\circ}\text{F}$ than the second sample.

17.

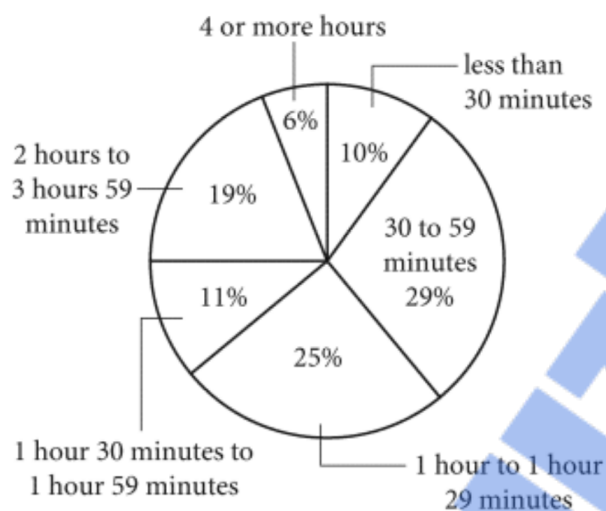


The box summarizes the distribution of 100 distinct values in a data set. Which of the following intervals contains the most data values?

- A) $10 \leq x \leq 40$
- B) $40 \leq x \leq 70$
- C) $70 \leq x \leq 90$
- D) $90 \leq x \leq 100$

18.

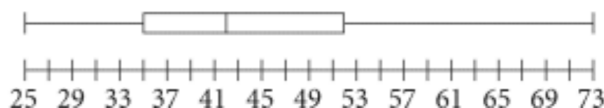
According to a study done by the Bureau of Labor Statistics, on a typical day between the years 2003 and 2006, about 16% of Americans aged 15 and older exercised. The pie chart below shows the distribution of the length of time those people spent exercising each day.



Based on the data shown, which of the following could be the median length of time spent exercising each day for those people who exercised?

- A) 0 minutes
- B) 45 minutes
- C) 1 hour 5 minutes
- D) 2 hours 10 minutes

19.



The daily high temperatures, in degrees Fahrenheit, for a city in the month of February in 2017 are summarized in the box plot shown. Which of the following is closest to the median of the high temperatures, in degrees Fahrenheit, of the city in February 2017?

- A) 35
- B) 42
- C) 48
- D) 51

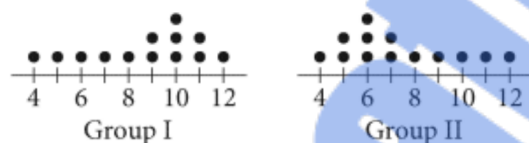
20.

List A	List B
1	2
2	3
3	3
4	4
5	4
6	5

The table above shows two lists of numbers. Which of the following is a true statement comparing list A and list B?

- A) The means are the same, and the standard deviations are different.
- B) The means are the same, and the standard deviations are the same.
- C) The means are different, and the standard deviations are different.
- D) The means are different, and the standard deviations are the same.

21.



The dot plots show 13 data values from group I and 13 data values from group II. Which of the following statements best compares the means and standard deviations of the values for each group?

- A) The mean of group I is greater than the mean of group II, and the standard deviation of group I is greater than the standard deviation of group II.
- B) The mean of group II is greater than the mean of group I, and the standard deviation of group II is greater than the standard deviation of group I.
- C) The mean of group I is greater than the mean of group II, but the standard deviations of group I and group II are the same.
- D) The mean of group II is greater than the mean of group I, but the standard deviations of group I and group II are the same.

22.

A sample of 200 bags of flour was selected at random from a production line, and the weights of the bags were analyzed. The mean weight was found to be 79.82 ounces (oz). The margin of error associated with the estimate is 0.24 oz. Based on the estimate and margin of error, which of the following is the most plausible conclusion about the mean weight of all bags of flour produced on the line?

- A) It is less than 79.82 oz.
- B) It is greater than 79.82 oz.
- C) It is less than 79.58 oz or greater than 80.06 oz.
- D) It is between 79.58 oz and 80.06 oz.

23.

A researcher asked study participants to report the number of jumps they observed in a video of a person jumping rope. The frequency table summarizes the number of jumps that each of the 30 participants reported.

Number of jumps reported	Frequency
0	1
1 to 10	0
11 to 20	2
21 to 30	17
31 to 40	10

Due to a conflict of interest, the participant who reported 0 jumps was removed from the study. When the participant who reported 0 jumps was removed from the study, what was the effect on the mean number of jumps reported?

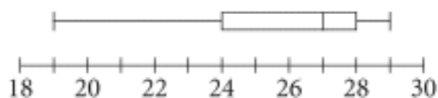
- A) The mean decreased.
- B) The mean increased.
- C) The mean remained the same.
- D) The effect on the mean cannot be determined.

24.

A total of 180 space walks, with a combined duration of 1,130 hours, were required to assemble the International Space Station. In which of the following equations does x represent the average (arithmetic mean) duration, in hours, of the 180 space walks?

- A) $1,130 = 180x$
- B) $180 = 1,130x$
- C) $x = (180)(1,130)$
- D) $x = 180 + 1,130$

25.



The box plot shown summarizes the values in a data set. What is the median value of the data set?

- A) 10
- B) 19
- C) 24
- D) 27

26.

The closing price, in dollars, per barrel of crude oil is listed in the table for 5 days in May 2016.

Date	Closing price per barrel (dollars)
May 2, 2016	47.69
May 3, 2016	46.99
May 4, 2016	46.90
May 5, 2016	47.25
May 6, 2016	47.78

What is the median closing price, in dollars, per barrel of crude oil during this 5-day period?

- A) \$46.90
- B) \$47.25
- C) \$47.32
- D) \$47.69

27.

The manager assigns the values shown below to each response choice to calculate a mean rating for the survey question.

Response	Rating
Strongly agree	2
Agree	1
Neither agree nor disagree	0
Disagree	-1
Strongly disagree	-2

What is the mean rating for all 50 responses?

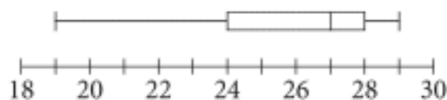
- A) 0.27
- B) 0.54
- C) 0.80
- D) 0.90

28. 137, 163, 115, 137, 179, 155, 171

A geologist recorded the heights, in feet, of a geyser's eruptions over a twelve-hour period. The heights are given in the list above. What is the median height recorded by the geologist?

- A) 115
- B) 137
- C) 151
- D) 155

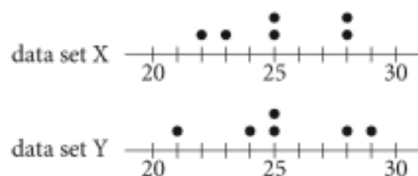
29.



The box plot shown summarizes the values in a data set. What is the median value of the data set?

- A) 10
- B) 19
- C) 24
- D) 27

30.



Data set X and data set Y are displayed by the two dot plots shown. Which of the following is(are) the same for both data sets?

- I. The mean
- II. The median

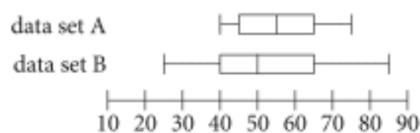
- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

31.

For a set of 35 integers, the mean is greater than the median. Which of the following CANNOT be true about the integers?

- A) They are all even.
- B) They are all odd.
- C) They are all different.
- D) They are consecutive.

32.



Two data sets are summarized by the box plots shown. Based on the box plots, which of the following must be true?

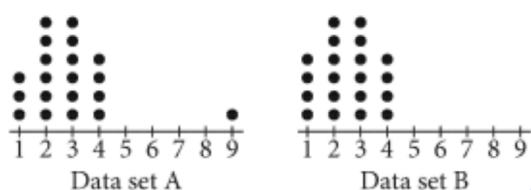
- A) The median of data set A is less than the median of data set B.
- B) The median of data set B is less than the median of data set A.
- C) There are more data values in data set A than in data set B.
- D) There are more data values in data set B than in data set A.

33.

A group of 10 students played a certain game. Every player received a score equal to an integer from 1 to 10, inclusive. For the 10 players, the mean score was 4. If more than half of the players received a score greater than 5, which of the following is true about the mean score of the remaining players?

- A) It must be less than 4.
- B) It must be equal to 4.
- C) It must be between 4 and 5.
- D) It must be greater than 5.

34.



Which statement best compares the means of the two data sets shown?

- A) The mean of data set A is greater than the mean of data set B.
- B) The mean of data set A is equal to the mean of data set B.
- C) The mean of data set A is less than the mean of data set B.
- D) There is not enough information to compare the means.

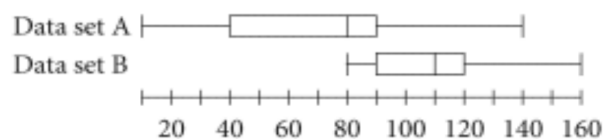
35.

Name	Time
Aaron	7 min 21 sec
Jamal	5 min 5 sec
Maya	5 min 21 sec
Aliyah	5 min 17 sec
Nacek	5 min 10 sec

Five students ran a mile, and their times are shown in the table above. If Aaron's time is removed from the data set, which of the following measures will change the least?

- A) Mean
- B) Median
- C) Maximum
- D) Range

36.

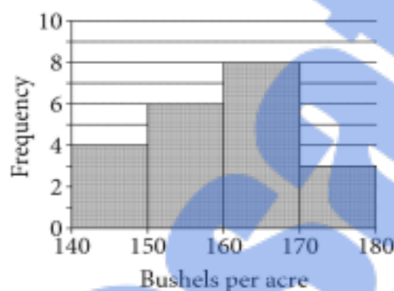


Each of the box plots shown summarizes a data set. Data set A has a range of 130, and data set B has a range of 80. If the two data sets are combined into one data set, what is the range of the combined data set?

- A) 50
- B) 105
- C) 150
- D) 210

37.

The histogram summarizes the distribution of corn production, in bushels per acre, over a 21-acre farm. The first bar represents the number of acres that produced at least 140 bushels but less than 150 bushels. The second bar represents the number of acres that produced at least 150 bushels but less than 160 bushels, and so on.

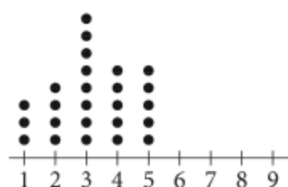


The median production, in bushels per acre, of these 21 acres is included in which interval?

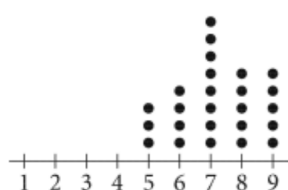
- A) At least 140 but less than 150
- B) At least 150 but less than 160
- C) At least 160 but less than 170
- D) At least 170 but less than 180

38.

Two different data sets are displayed in the dot plots shown. Which of the following statements is true?



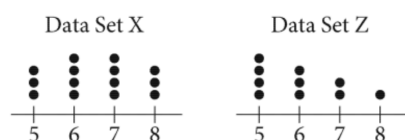
Data set I



Data set II

- A) The mean of data set I is less than the mean of data set II, but the standard deviations of the data sets are the same.
- B) The mean of data set I is less than the mean of data set II, but the standard deviation of data set I is greater than the standard deviation of data set II.
- C) The mean and standard deviation of data set I are less than the mean and standard deviation of data set II.
- D) The mean and standard deviation of data set I are the same as the mean and standard deviation of data set II.

39.



The dot plots above show the distribution of the values of two data sets. Which of the following is(are) true?

- I. The median of the values in data set X is greater than the median of the values in data set Z.
 - II. The mean of the values in data set X equals the mean of the values in data set Z.
- A) Neither I nor II
 - B) I only
 - C) II only
 - D) Both I and II

40.

Data set A consists of positive values, with no two values the same. Data set B is formed by removing the largest value from data set A. Which of the following statements must be true?

- I. The mean of data set B is smaller than the mean of data set A.
- II. The range of data set B is smaller than the range of data set A.

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

41.

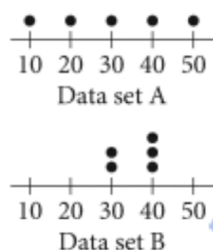
Each dot plot shown represents a data set.



Which of the following statements correctly compares the means of the two data sets?

- A) The mean of data set A is greater than the mean of data set B.
- B) The mean of data set A is less than the mean of data set B.
- C) The means of the data sets are equal.
- D) There is not enough information to compare the means.

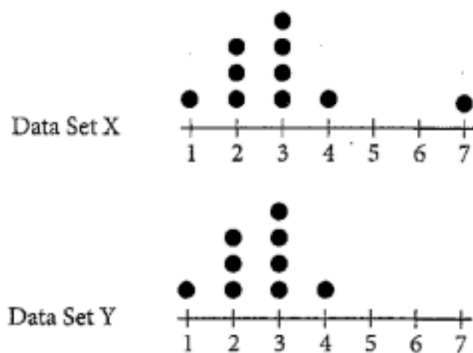
42.



Which of the given data sets has a greater standard deviation?

- A) Data set A
- B) Data set B
- C) The standard deviations of data set A and data set B are equal.
- D) There is not enough information to determine which data set has a greater standard deviation.

43.



Data set X and data set Y are represented by the dot plots shown. Which of the following is (are) true?

- I. The median of data set X is greater than the median of data set Y.
- II. The mean of data set X is greater than the mean of data set Y.

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

44.

Class size	Number of classes
23	1
28	2
29	6
30	6
31	5

The table above shows the distribution of class sizes for the 20 math classes at a high school.

What is the mean math class size?

- A) 28.2
- B) 28.5
- C) 29.4
- D) 30.0